

ARCHIT SHUKLA

Boulder, CO | +1 303 931 8281 | Archit.Shukla@colorado.edu | Portfolio

EDUCATION

- University Of Colorado Boulder** • Boulder, CO Aug 2024 - May 2026
Masters of Science • Computer Science • GPA: 3.91/4
Relevant Coursework: Design and Analysis of Algorithms, Operating systems, Big Data Architecture, Generative AI for Software Development, Machine Learning, Computer Vision, User-Centered Design and Development
- SRM Institute of Science & Technology** • Chennai, India Aug 2020 - May 2024
Bachelor of Technology • Computer Science & Engineering • GPA: 3.81/4

TECHNICAL SKILLS

- Languages: **Python, Java, C/C++, JavaScript (ES6+), Go, SQL**
- Backend: **Microservices, REST APIs, Spring Boot, Flask/FastAPI, Node.js, Async & Event-Driven Systems**
- Databases: **PostgreSQL, MongoDB, DynamoDB, ETL Pipelines, Query Optimization, Vector DB (Qdrant)**
- Cloud & DevOps: **Docker, Kubernetes, CI/CD, Linux, Git, GCP, AWS (EC2, S3, IAM, ECS, SQS)**
- AI Systems: **Agent-Based Systems, Multi-Agent Architectures, Reinforcement Learning (PPO), RAG Pipelines, LangChain, LLM Integration (OpenAI/Bedrock), Embeddings, Model Evaluation**

WORK EXPERIENCE

- Peleg Lab, BioFrontiers Institute** Aug 2025 – Jan 2026
Graduate Researcher—Distributed Systems Engineering Boulder, CO
- Increased evaluation throughput by **35%** and eliminated hundreds of GPU-hours per cycle by architecting **multi-threaded parallel pipelines in Python/C++ on GCP**.
 - Designed and provisioned **containerized, GPU-backed execution environments**, ensuring reproducible large-scale experiments and stable long-running workloads across heterogeneous systems.
 - Reduced failure diagnosis time by **40%** via GDB/Valgrind debugging and CI/CD integration, improving pipeline reliability.
- CodeClause** June 2024 – July 2024
Software Engineering Intern—Backend & Data Systems Bangalore, India
- Increased backend throughput by **20%** and reduced production latency by optimizing **Flask-based REST APIs and SQL queries** in live, high-traffic environments serving concurrent user requests.
 - Reduced release failures by **25%** by streamlining **build, test, and deployment pipelines** using Jenkins CI/CD and GitHub, improving release consistency and deployment reliability.
 - Decreased issue diagnosis time by **35%** by implementing structured **unit testing, logging, and monitoring**.
- Adflix Media** Apr 2023 – Sep 2023
Software Engineer Intern—Big Data Infrastructure Pune, India
- Processed **5M+ records daily** by scaling **SQL-driven ETL pipelines** using Dockerized microservices, maintaining low operational overhead and consistent batch execution performance.
 - Reduced issue detection time by **30%** by developing Grafana observability dashboards for latency and throughput monitoring across distributed data pipelines.
 - Cut recovery time by **25%** through implementation of structured logging and cloud monitoring, improving production visibility and system transparency.
- University of Colorado Boulder** Jan 2026 – Present
Teaching Assistant—Lab Systems Boulder, CO
- Directed end-to-end execution of **3 lab sections** (53 students), resolving live failures under time constraints.
 - Analyzed workflow data in **R** to identify recurring failure patterns and implement operational improvements across sections.

PROJECTS

- RepoGraph V2 – Distributed Debugging Platform** | *Backend Services + Static Analysis*[Paper] Aug 2025 – Dec 2025
- Implemented a **distributed backend system in Python on Linux/UNIX** to localize, patch, and verify flaky tests across **100K+ LOC** codebases by analyzing execution order, shared state, and dependency graphs at scale.
 - Architected **graph-based dependency closures** and exposed results via **versioned RESTful services**, enabling coordinated, fault-tolerant analysis, patching, and verification workflows across components.
 - Reduced inter-service data transfer by **70–85%** using graph-scoped retrieval and optimized serialization strategies, improving throughput and stability under concurrent, parallel workloads.
- Real-Time Anomaly Detection Service** | *Streaming Pipelines + Backend Services* [Demo] July 2025 – Aug 2025
- Engineered a **cloud-native streaming pipeline** to ingest, validate, and preprocess continuous event data, processing **2K+ records per minute** with predictable, low-latency guarantees.
 - Exposed anomaly predictions via a **backend REST API**, improving detection accuracy by **30%** over baseline models through optimized feature preprocessing and inference logic.
 - Deployed **containerized, auto-scaling infrastructure** using Docker and CI/CD on GCP, ensuring fault isolation, zero-downtime rollouts, and reliable production execution.
- LoanSphere – Backend Data Processing Platform** | *APIs + Data Pipelines* [Demo] Jan 2025 – May 2025
- Engineered **SQL-driven backend pipelines** processing **17M+ structured records** with scalable cleaning, normalization, and feature encoding across high-volume transactional datasets.
 - Implemented **Flask/FastAPI and Node.js services** with versioned **REST APIs**, reducing validation errors by **30%** while maintaining bounded response times under concurrent user requests.
 - Streamlined containerized **CI/CD workflows** to deploy backend services with reproducible builds, environment consistency, and reduced manual release overhead.